UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PROGRAMMING I
COURSE CODE	:	CSC402
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC402 mock paper (November 2026). Generated answers show working; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 10 x 1 mark; Part B = 5 questions (60 marks); Part C = 2 x 15 marks; total = 100 marks.

4. Questions: 17. Slots: 22. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 21 unique CSC402 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — ANSWER KEY

Q	Ans	Question	Solution
A1	B	Which structure chooses between two alternative paths?	if...else performs two-way selection.
A2	C	What is displayed by the following code?	The loop displays 2, 4, 6 and 8 without spaces.
A3	D	What does the % operator return?	% is the modulus operator and returns the remainder.
A4	A	Which phase of the program development life cycle defines what the program must do?	Analysis establishes the requirements before design and coding.
A5	D	Which of the following declarations are valid?	a and c are valid array declarations.
A6	A	Which input pattern stops when the user enters -1?	A sentinel value ends the loop.
A7	B	Which term describes a finite sequence of precise steps that solves a problem?	An algorithm is a finite, ordered set of unambiguous steps.
A8	C	Which data type stores a whole number?	int stores whole numbers; float stores real values.
A9	D	Which of the following are selection structures?	if and switch select among alternatives.
A10	A	Which declaration creates a floating-point variable named price?	float is used for real numbers.

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PART B — WORKED ANSWERS

QUESTION 1

Topic: debugging testing

a) Define algorithm and pseudocode.

An algorithm is a finite sequence of precise steps that solves a problem. Pseudocode is a structured English-like description of an algorithm.

Marking: Award 1 mark for each correct definition.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

Syntax error: missing semicolon. Runtime error: division by zero. Logic error: using < instead of <= in a condition.

Marking: Award 0.5 mark for each type and 0.5 mark for a valid example.

(3 marks)

QUESTION 2

Topic: fundamentals

a) Draw a flowchart for the grade classification.

Start -> read score -> decision score >= 80 -> A; else score >= 60 -> B; else score >= 50 -> C; else score >= 40 -> D; else F -> display grade -> end.

Marking: Award marks for terminal symbols, input, four/five decisions, each output and correct flow.

(5 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

IF score >= 80 THEN grade = A ELSE IF score >= 60 THEN grade = B ... ENDIF; use if/else if in C++ and display grade.

Marking: Award marks for correct pseudocode, complete C++ structure, conditions, input and output.

(8 marks)

QUESTION 3

Topic: data types

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

int age; double price; const double DISCOUNT = 0.10; char grade;

Marking: Award marks for correct types, names and const initialisation.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

cin >> age >> price; price = price - price * DISCOUNT; cout << price;

Marking: Award marks for input, discount expression and output.

(5 marks)

QUESTION 4

Topic: input output

a) Trace the program and show the exact output.

1\n12\n123

Marking: Award marks for each line and the exact row lengths.

(7 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

The inner loop must initialise j = 1 for every row and the outer loop must run to 4. Correct code: for (int i=1;i<=4;i++){for(int j=1;j<=i;j++) cout<<j; cout<<endl;}

Marking: Award marks for each error and the corrected loop.

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(8 marks)

QUESTION 5

Topic: arrays

a) Declare an array named prices using the values in Table 1. Display all values.

double prices[5] = {20.5, 55.0, 15.9, 75.0, 10.0}; for (int i=0;i<5;i++) cout << prices[i] << endl;

Marking: Award marks for correct type, initialiser, loop and output.

(6 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

double sale[5]; for(int i=0;i<5;i++){ sale[i] = prices[i] > 50 ? prices[i]*0.9 : prices[i]; } then display both arrays.

Marking: Award marks for second array, condition, discount, preservation and output.

(10 marks)

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PART C — WORKED ANSWERS

QUESTION 1

A catering company charges the meal prices shown in Table 1.

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;
int main() {
    string name;
    int quantity, grandQuantity = 0;
    double unitPrice, total, grandTotal = 0;
    char again;
    do {
        cout << "Customer name: ";
        getline(cin >> ws, name);
        cout << "Quantity: ";
        cin >> quantity;
        if (quantity < 0) {
            cout << "Invalid quantity" << endl;
        } else {
            if (quantity >= 200) unitPrice = 9.50;
            else if (quantity >= 100) unitPrice = 10.50;
            else if (quantity >= 50) unitPrice = 11.50;
            else unitPrice = 12.50;
            total = quantity * unitPrice;
            grandTotal += total;
            grandQuantity += quantity;
            cout << fixed << setprecision(2);
            cout << "Receipt for " << name << ": "
                 << quantity << " x RM" << unitPrice
                 << " = RM" << total << endl;
        }
        cout << "Another customer? (y/n): ";
        cin >> again;
    } while (again == 'y' || again == 'Y');
    cout << "Total quantity: " << grandQuantity << endl;
    cout << "Total sales: RM" << fixed << setprecision(2)
         << grandTotal << endl;
    return 0;
}
```

Marking: Award 3 marks for input/validation, 6 for price selection and total, 3 for repetition, 2 for final totals and 1 for output clarity.

(15 marks)

QUESTION 2

A cafe sells the items shown in Table 1. A 10% discount is given when the subtotal is at least RM50.

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;
int main() {
    int code, quantity;
    double price = 0, subtotal, discount = 0;
    string name;
    cout << "Item code (1 coffee, 2 tea, 3 juice): ";
    cin >> code;
    cout << "Quantity: ";
```

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```
    cin >> quantity;
    if (code == 1) { name = "Coffee"; price = 8.50; }
    else if (code == 2) { name = "Tea"; price = 6.00; }
    else if (code == 3) { name = "Juice"; price = 9.50; }
    else { cout << "Invalid item code" << endl; return 1; }
    subtotal = price * quantity;
    if (subtotal >= 50) discount = subtotal * 0.10;
    cout << fixed << setprecision(2);
    cout << "Item: " << name << endl;
    cout << "Quantity: " << quantity << endl;
    cout << "Subtotal: RM" << subtotal << endl;
    cout << "Discount: RM" << discount << endl;
    cout << "Amount to pay: RM" << subtotal - discount << endl;
    return 0;
}
```

Marking: Award 3 marks for input, 5 for price/subtotal, 4 for discount, 2 for formatting and 1 for completeness.

(15 marks)

END OF MARKING SCHEME

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